# Requirements Traceability Matrix

Deliverable 3, Campus Ride-Pooling and Split-Fare System

Prepared by Mainak Sarkar  Requirement → Jira → GitHub → test  Tests run on 2026-11-06

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**Result.** 194 requirements of the Deliverable 1 specification are traced to 80 Jira issues, 51 merged pull requests, 186 automated test cases and 117 test groups in 116 test files. Every one of those passed in the recorded runs: the unit suite (1544 tests) and the integration and adversarial suite (112 tests against a real PostgreSQL database), both on the master branch.

## 1. Reading the matrix

| | |
|---|---|
| **Requirement** | The requirement's identifier and statement from the Deliverable 1 specification (rider, ride owner and Operations Admin documents). Long statements are shortened; the full text of each is in `traceability-matrix.csv`. |
| **Jira** | The CS455 Jira issues of the pull requests in the GitHub column, as named in their titles, branches, own commits and Jira sections. |
| **GitHub** | The merged pull requests that introduced the code under test, that cite the requirement, and that introduced its tests; up to five are shown, the CSV lists all of them. |
| **Tests** | The test files that verify the requirement, each with the number of its test cases and test groups that do, largest first. A test verifies a requirement when its own title cites the requirement's identifier. An identifier in the title of a `describe()` block is about the block as a whole, so it links that block as one test group, which passes when every test in it passed; it does not link each test inside. Requirements marked † are verified by tests whose names do not cite the identifier; each of those links was made by reading the test and is listed in `data/manual-links.json`. |
| **Result** | Passed test cases and test groups out of those linked, from the recorded runs. |

## 2. How the matrix is built

Every cell comes from a recorded source, so the matrix can be rebuilt and checked rather than maintained by hand. A requirement is listed when at least one passing automated test verifies it.

- **Requirements**: parsed from the Deliverable 1 requirement documents in `Deliverables/D1/Requirements Engineering/users/`.
- **Tests**: the recorded unit run (`data/unit-cases.json`) and integration run (`Deliverables/D3/test-run/integration-results.json`). Titles such as "complaints (FR-RD-13.1 to 13.4)" or "(OA-FR-34, 36)" cite every requirement in the range or list. The build refuses to run if a link in `data/manual-links.json` matches no recorded test.
- **Pull request text**: HTML comments in a pull request body are the template's hints and examples, so they are removed before looking for requirement identifiers and Jira keys.
- **GitHub**: a snapshot of the 98 merged pull requests with their changed files and commits (`data/prs.json`). The pull request that introduced a file is the first merged one that changed it. A commit counts only for the first merged pull request that contains it, since a stacked pull request also lists the commits below it.

- **Coverage**: NFR-RD-17 is verified by the unit suite's statement coverage (87.7 %) against the 80 % gate that continuous integration enforces.

```
node Deliverables/D3/Traceability_Matrix/snapshot-prs.mjs                                  (merged pull requests)
node Deliverables/D3/Traceability_Matrix/make-matrix.mjs --unit <run.json> --coverage <summary.json>   (tables and CSV)
```

The commands that record the test runs are in `Deliverables/D3/Traceability_Matrix/README.md`.

## 3. Summary by area

| Area | Requirements | Jira issues | Pull requests | Tests and groups |
|---|---|---|---|---|
| Rider functional requirements | 81 | 70 | 41 | 175 |
| Rider AI agent requirements | 13 | 17 | 5 | 28 |
| Rider non-functional requirements | 7 | 28 | 18 | 18 |
| Ride Owner functional requirements | 39 | 50 | 25 | 66 |
| Ride Owner non-functional requirements | 7 | 32 | 14 | 15 |
| Operations Admin functional requirements | 16 | 20 | 12 | 33 |
| System functional requirements (Operations Admin) | 21 | 44 | 23 | 29 |
| System non-functional requirements (Operations Admin) | 10 | 23 | 13 | 16 |
| **Total (distinct)** | **194** | **80** | **51** | **303** |

## 4. The matrix

| Requirement | Jira | GitHub | Tests | Result |
|---|---|---|---|---|
| **Rider functional requirements** | | | | |
| **FR-RD-01.1**† When a user registers, the system shall accept only email addresses whose domain is exactly iitk.ac.in (P-01) and shall reject any other address with a message that … | CS455-19, CS455-130 | #20, #22, #95 | `components/PageShells` (1), `lib/auth` (1) | Pass 2/2 |
| **FR-RD-01.3**† The system shall not allow an account whose email address is unverified to use any rider function other than verification. | CS455-19 | #20, #22 | `api/auth/login/route` (1) | Pass 1/1 |
| **FR-RD-01.4**† The system shall end a rider's session after 7 days without activity (P-04) and immediately when the rider signs out. | CS455-19, CS455-86, CS455-89 | #20, #22, #67, #69 | `lib/auth` (2) | Pass 2/2 |
| **FR-RD-01.5** If an administrator suspends a rider's account, then the system shall from that moment reject the rider's new join requests, reconfirmations, pool chat messages and … | CS455-21, CS455-22, CS455-23, CS455-33, CS455-34, +14 | #24, #38, #41, #47, #67, +4 | `rides/[id]/ratings/route` (1), `api/rides/route` (1), `lib/enforcement-rules` (1), +5 files | Pass 8/8 |
| **FR-RD-01.6** When an administrator issues a warning to a rider, the system shall show the warning and its reason category at the rider's next sign-in and require the rider to … | CS455-21, CS455-22, CS455-23, CS455-39, CS455-42, +8 | #24, #38, #47, #67, #71, +2 | `admin/complaints-and-enforcement` (1) | Pass 1/1 |
| **FR-RD-02.1**† When a rider signs in for the first time, the system shall require a display name of 2--40 characters and a default pickup point chosen from the configured list, and … | CS455-12, CS455-15, CS455-31, CS455-86, CS455-89, +1 | #17, #33, #67, #69, #95 | `lib/profile-data` (1), `lib/session` (1) | Pass 2/2 |
| **FR-RD-02.2**† The system shall allow the rider to select up to 10 (P-05) interest tags only from the administrator-maintained tag list, and shall not accept free-text tags. | CS455-31 | #33 | `api/profile/route` (1) | Pass 1/1 |

† verified by tests whose names do not cite the identifier (see `data/manual-links.json`)

| Requirement | Jira | GitHub | Tests | Result |
|---|---|---|---|---|
| **FR-RD-02.3**† The system shall allow the rider to edit their display name, default pickup point, tags and preferences at any time; changes shall apply to later searches and … | CS455-31 | #33 | `api/profile/route` (1), `profile/ProfileEditor` (1) | Pass 2/2 |
| **FR-RD-02.4** The system shall show other users only a rider's display name, the tags the rider has marked visible, the aggregate rating (subject to FR-RD-12.3) and the number of … | CS455-27, CS455-28, CS455-29, CS455-31, CS455-39, +12 | #27, #28, #33, #38, #75, +2 | `lib/ride-view` (3), `rides/RideDetails` (1), `ai/chatbot-tools` (1), +2 files | Pass 7/7 |
| **FR-RD-02.5** The system shall allow the rider to save default preferences --- preferred vehicle type and maximum acceptable fare share --- which pre-fill search filters. | CS455-20, CS455-33, CS455-34, CS455-35, CS455-36, +1 | #41, #45 | `rides/SearchRideForm` (1), `lib/ride-search` (1) | Pass 2/2 |
| **FR-RD-02.7** The system shall allow the rider to add an optional 10-digit Indian mobile number, visible only to administrators, included in any SOS incident the rider raises … | CS455-85, CS455-112 | #64, #86 | `trip/sos` (1) | Pass 1/1 |
| **FR-RD-03.1** The system shall allow the rider to search rides by destination and date (both required), a departure time window, a pickup point (default: the profile's default) … | CS455-20, CS455-21, CS455-22, CS455-23, CS455-33, +5 | #24, #41, #45, #85 | `lib/ride-search-rules` (1), `rides/offer-and-search` (1) | Pass 2/2 |
| **FR-RD-03.2** The system shall return only rides that are open, have a lock time (P-06) in the future, whose owner is not suspended, and that either have at least one available … | CS455-21, CS455-22, CS455-23, CS455-33, CS455-34, +12 | #24, #41, #75, #85 | `rides/offer-and-search` (3), `ai/chatbot-tools` (1) | Pass 4/4 |
| **FR-RD-03.3** For each result the system shall display the destination, departure time, vehicle type, available seats, total fare, the rider's estimated share (FR-RD-08.1), the … | CS455-21, CS455-22, CS455-23, CS455-27, CS455-28, +7 | #24, #28, #41, #85 | `rides/RideDetails` (1), `rides/offer-and-search` (1) | Pass 2/2 |
| **FR-RD-03.6** The ride details view shall also show the public profiles (FR-RD-02.4) of the owner and accepted riders, the pickup order with each occupant's pickup point, and the … | CS455-27, CS455-28, CS455-29, CS455-100, CS455-101, +6 | #28, #75 | `rides/RideDetails` (1), `ai/chatbot-tools` (1) | Pass 2/2 |
| **FR-RD-04.1**† When the rider selects "Ask AI to Suggest" for a search, the system shall return at most 5 (P-11) open rides ranked by suitability for that search, each with up to … | CS455-37, CS455-38 | #51 | `rides/AskAiRecommendations` (1), `ai/validate-ranking` (1) | Pass 2/2 |
| **FR-RD-04.2**† The system shall base rankings and pros/cons only on: estimated share, estimated pickup time relative to the rider's time window, the change the rider would cause … | CS455-37, CS455-38 | #51, #92, #93 | `ai/shared-interest-enrichment` (1), `ai/shared-interest-tags` (1), `ai/validate-ranking` (1) | Pass 3/3 |
| **FR-RD-04.3**† Before displaying AI suggestions, the system shall verify that every suggested ride exists, is open and has at least one available seat, and that every fare, time … | CS455-37, CS455-38, CS455-130 | #51, #95 | `ai/recommendation-pipeline` (1), `ai/validate-ranking` (1) | Pass 2/2 |
| **FR-RD-04.5**† If the AI service returns an error or no valid response within 8 seconds (P-12), or none of its suggestions passes the checks of FR-RD-04.3, the system shall … | CS455-37, CS455-38 | #51 | `ai/fallback-ranking` (2), `ai/recommendation-pipeline` (1) | Pass 3/3 |
| **FR-RD-05.1** When the rider sends a natural-language query about rides, the system shall convert it into search filters, show the interpreted filters as editable values, and … | CS455-100, CS455-101, CS455-102, CS455-103, CS455-104, +4 | #75, #76 | `chatbot/ChatbotInterface` (1), `ai/chatbot` (1) | Pass 2/2 |
| **FR-RD-05.2** When asked about an owner or co-rider, the chatbot shall disclose only the public profile fields of FR-RD-02.4. | CS455-100, CS455-101, CS455-102, CS455-103, CS455-104, +3 | #75 | `ai/chatbot-tools` (1), `ai/chatbot` (1) | Pass 2/2 |
| **FR-RD-05.3** When the rider asks the chatbot to change a preference or tag, the system shall show the proposed change and apply it only if the rider confirms it; an unconfirmed … | CS455-100, CS455-101, CS455-102, CS455-103, CS455-104, +4 | #75, #76 | `ai/chatbot-tools` (1), `ai/chatbot` (1), `ai/preference-proposals` (1) | Pass 3/3 |
| **FR-RD-05.4** The chatbot shall not submit, withdraw or reconfirm join requests, record payments, submit ratings or complaints, post pool chat messages, or raise or cancel SOS … | CS455-100, CS455-101, CS455-102, CS455-103, CS455-104, +4 | #75, #76 | `api/chatbot/route` (1), `chatbot/ChatbotInterface` (1), `ai/chatbot` (1) | Pass 3/3 |
| **FR-RD-05.5** The chatbot shall decline requests unrelated to ride pooling on the platform with a short scope message. | CS455-100, CS455-101, CS455-102, CS455-103, CS455-104, +3 | #75 | `ai/chatbot` (1) | Pass 1/1 |
| **FR-RD-05.6** The system shall keep chatbot conversation content only in the browser tab where it takes place, discarding it when the tab is closed or reloaded or the rider signs … | CS455-100, CS455-101, CS455-102, CS455-103, CS455-104, +4 | #75, #76 | `chatbot/ChatbotInterface` (3), `ai/chatbot-audit` (1), `ai/chatbot-session` (1) | Pass 5/5 |

† verified by tests whose names do not cite the identifier (see `data/manual-links.json`)

| Requirement | Jira | GitHub | Tests | Result |
|---|---|---|---|---|
| **FR-RD-05.7** If the AI service is unavailable, the chatbot shall say so and link to manual search. | CS455-100, CS455-101, CS455-102, CS455-103, CS455-104, +4 | #75, #76 | `api/chatbot/route` (1), `chatbot/ChatbotInterface` (1), `ai/chatbot` (1) | Pass 3/3 |
| **FR-RD-06.1** When a rider submits a join request for a ride and a pickup point, the system shall create the request in state Pending, store a snapshot of the ride's departure … | CS455-33, CS455-34, CS455-35, CS455-36, CS455-45, +3 | #41, #55, #85 | `rides/join-requests` (1) | Pass 1/1 |
| **FR-RD-06.2** If, at submission, the ride is not open, has no available seat or is past its lock time; the rider already has an active or accepted request on it; the ride departs … | CS455-33, CS455-34, CS455-35, CS455-36, CS455-45, +4 | #41, #55, #67, #85 | `rides/join-requests` (1), `rides/seats-and-lock` (1) | Pass 2/2 |
| **FR-RD-06.3** When the ride owner accepts a request, the system shall, as one atomic operation, re-check that the ride is open with at least one available seat, that the request … | CS455-21, CS455-22, CS455-23, CS455-76, CS455-77, +4 | #24, #55, #64, #93, #95, +2 | `adversarial/races` (1), `rides/seats-and-lock` (1) | Pass 2/2 |
| **FR-RD-06.4** When a request is accepted or rejected, the system shall notify the rider within 2 seconds (P-13), and an accepted ride shall appear in the rider's upcoming rides. | CS455-12, CS455-15, CS455-27, CS455-28, CS455-29, +18 | #17, #27, #29, #41, #55, +2 | `lib/ride-view` (1), `rides/join-requests` (1) | Pass 2/2 |
| **FR-RD-06.5** If, while a request is Pending, the ride's departure time or the rider's estimated pickup time moves by more than 10 minutes (P-09), or the rider's estimated share … | CS455-76, CS455-77, CS455-86, CS455-111 | #55, #67, #85 | `rides/manage-ride` (1) | Pass 1/1 |
| **FR-RD-06.6**† When the rider reconfirms a request, the system shall replace its snapshot with current values and return it to Pending, keeping its original submission time for … | CS455-76, CS455-77, CS455-86 | #55, #67 | `lib/join-requests` (2) | Pass 2/2 |
| **FR-RD-06.7** The system shall mark an active request Expired when the ride reaches its lock time, and Ride Cancelled when the owner cancels the ride, notifying the rider in both … | CS455-76, CS455-77, CS455-111 | #55, #85 | `rides/manage-ride` (1), `rides/seats-and-lock` (1) | Pass 2/2 |
| **FR-RD-06.8** When a request is accepted, the system shall withdraw the rider's other active requests for rides departing within 2 hours (P-08) of the accepted ride and notify … | CS455-76, CS455-77, CS455-86, CS455-111 | #55, #67, #85 | `rides/seats-and-lock` (1) | Pass 1/1 |
| **FR-RD-06.9** The system shall treat repeated submissions carrying the same idempotency key within 24 hours (P-20) as one request. | CS455-33, CS455-34, CS455-35, CS455-36, CS455-45, +5 | #41, #55, #64, #85, #87, +1 | `adversarial/replays-and-state` (2), `bugs/idempotency-key-reuse` (1), `rides/join-requests` (1) | Pass 4/4 |
| **FR-RD-06.10** The system shall show the rider the current state, state history and reason codes of each of their requests. | CS455-12, CS455-15, CS455-33, CS455-34, CS455-35, +5 | #17, #41, #55, #85 | `rides/join-requests` (2) | Pass 2/2 |
| **FR-RD-06.11** When the owner cancels a ride on which the rider holds an accepted seat, before or after lock time but before the trip starts and before the departure window ends, … | CS455-76, CS455-77, CS455-86, CS455-111 | #55, #67, #85 | `rides/manage-ride` (1) | Pass 1/1 |
| **FR-RD-07.1** The system shall allow the rider to withdraw an active request at any time; the request becomes Withdrawn and the owner is notified. | CS455-33, CS455-34, CS455-35, CS455-36, CS455-45, +4 | #41, #55, #85, #95 | `rides/join-requests` (1) | Pass 1/1 |
| **FR-RD-07.2** When an accepted rider leaves a ride before its lock time, the system shall, as one atomic operation, mark the request Left and increase the available seats by one; … | CS455-33, CS455-34, CS455-35, CS455-36, CS455-45, +4 | #41, #55, #85, #93, #95 | `rides/join-requests` (1) | Pass 1/1 |
| **FR-RD-07.3** When an accepted rider leaves after the lock time and before the trip starts (trip state Scheduled; glossary: trip start), the system shall require explicit … | CS455-46, CS455-47, CS455-60, CS455-61, CS455-62, +7 | #42, #55, #57, #67, #93, +1 | `lib/complaint-rules` (1), `lib/trip` (1), `rides/seats-and-lock` (1) | Pass 3/3 |
| **FR-RD-07.5** The system shall not allow a rider to leave a trip whose state is Pickup in Progress or later (FR-RD-10.1); SOS shall remain available (UR-RD-15). | CS455-60, CS455-61, CS455-62, CS455-63, CS455-64, +5 | #55, #56, #64, #88 | `lib/trip-rules` (1), `adversarial/replays-and-state` (1) | Pass 2/2 |
| **FR-RD-08.1** Before a request is submitted, the system shall show the rider's estimated share as the ride's total fare divided by (current occupants + 1), rounded up to the next … | CS455-12, CS455-16, CS455-21, CS455-22, CS455-23, +29 | #18, #24, #28, #41, #75, +6 | `rides/RideDetails` (1), `ai/chatbot-tools` (1), `lib/fare` (1), +2 files | Pass 5/5 |
| **FR-RD-08.3** At lock time the system shall fix each occupant's share and shall not change it afterwards, except as FR-RD-07.3 provides. | CS455-12, CS455-16, CS455-60, CS455-61, CS455-62, +6 | #18, #55, #57, #85 | `lib/trip` (1), `rides/seats-and-lock` (1) | Pass 2/2 |

† verified by tests whose names do not cite the identifier (see `data/manual-links.json`)

| Requirement | Jira | GitHub | Tests | Result |
|---|---|---|---|---|
| **FR-RD-08.4** The system shall compute shares in whole rupees such that they sum exactly to the ride's total fare, using the rule in Table T-2. | CS455-12, CS455-16, CS455-20, CS455-21, CS455-24, +3 | #18, #21, #27 | `lib/ride-view` (1) | Pass 1/1 |
| **FR-RD-09.1**† When a ride locks, the system shall open a pool chat for the owner and accepted riders and notify each member. | CS455-12, CS455-16, CS455-33, CS455-34, CS455-35, +3 | #18, #41, #95 | `lib/ride-chat` (1) | Pass 1/1 |
| **FR-RD-09.2**† The system shall deliver each chat message to all connected members within 1 second (NFR-RD-03) and persist it, so that members who were offline see it when they ... | CS455-33, CS455-34, CS455-35, CS455-36, CS455-45, +1 | #41, #70 | `lib/ride-chat-realtime` (1) | Pass 1/1 |
| **FR-RD-09.3**† Only current members of a pool shall be able to read or post in its chat; a rider who leaves loses access immediately. | CS455-33, CS455-34, CS455-35, CS455-36, CS455-45 | #41 | `lib/ride-chat` (1) | Pass 1/1 |
| **FR-RD-09.4**† The system shall accept only text messages of at most 500 characters, and at most 20 messages per minute per member (P-14). | CS455-33, CS455-34, CS455-35, CS455-36, CS455-45 | #41 | `lib/ride-chat` (1) | Pass 1/1 |
| **FR-RD-09.5** The system shall make a pool chat read-only 24 hours after trip completion or ride cancellation and delete it 30 days after that time (P-15), unless an unresolved ... | CS455-12, CS455-16, CS455-33, CS455-34, CS455-35, +3 | #18, #41, #67 | `lib/ride-chat` (1) | Pass 1/1 |
| **FR-RD-09.6** The system shall allow a member to report a chat message, which opens a complaint (FR-RD-13.1) pre-filled with a reference to the message. | CS455-33, CS455-34, CS455-35, CS455-36, CS455-45, +4 | #41, #47, #49 | `complaints/ComplaintForm` (1), `lib/complaint-queries` (1), `lib/complaints` (1) | Pass 3/3 |
| **FR-RD-09.7**† The system shall offer a one-tap "I am at the pickup point" message. | CS455-33, CS455-34, CS455-35, CS455-36, CS455-45 | #41 | `rides/RideChat` (1) | Pass 1/1 |
| **FR-RD-10.1** The system shall show the rider the trip state --- Scheduled, Pickup in Progress, In Transit, Completed or Cancelled --- as set by the ride owner. | CS455-27, CS455-28, CS455-29, CS455-39, CS455-41, +8 | #27, #28, #37, #56, #57, +2 | `lib/ride-status` (1), `lib/trip-rules` (1), `lib/trip` (1), +1 files | Pass 4/4 |
| **FR-RD-10.2** The system shall show the rider the pickup order, each occupant's pickup point and the rider's own estimated pickup time, updated whenever the itinerary changes. | CS455-27, CS455-28, CS455-60, CS455-61, CS455-62, +4 | #27, #56, #64 | `lib/ride-view` (1), `lib/trip-itinerary` (1) | Pass 2/2 |
| **FR-RD-10.3** When the owner marks the rider as a no-show --- permitted only after the grace period of P-19 --- the system shall notify the rider and record the no-show; the rider ... | CS455-21, CS455-22, CS455-23, CS455-39, CS455-42, +14 | #24, #38, #47, #57, #71, +3 | `rides/TripPanel` (1), `lib/trip` (1), `admin/complaints-and-enforcement` (1), +1 files | Pass 4/4 |
| **FR-RD-10.4** The system shall allow the rider to check in at their pickup point, visible to the pool. | CS455-39, CS455-41, CS455-60, CS455-61, CS455-62, +5 | #37, #56, #57, #64, #86 | `rides/[id]/check-in/route` (1), `rides/TripPanel` (1), `lib/trip-rules` (1), +2 files | Pass 5/5 |
| **FR-RD-11.1**† When a trip is completed, the system shall set each rider's locked share to Due and show the amount payable to the owner. | CS455-39, CS455-41, CS455-60, CS455-61, CS455-62, +8 | #37, #57, #67, #69, #71, +1 | `lib/payment-settlement` (1), `trip/trip-flow` (1) | Pass 2/2 |
| **FR-RD-11.2**† The system shall allow the rider to mark a Due share as paid, with the method (UPI or cash) and an optional reference of up to 50 characters; the share becomes ... | CS455-12, CS455-16, CS455-86, CS455-89, CS455-94, +2 | #18, #67, #69, #71, #95 | `rides/SettlementPanel` (1), `lib/payment-rules` (1), `lib/payment-settlement` (1) | Pass 3/3 |
| **FR-RD-11.3**† When the owner confirms receipt, the share shall become Confirmed; when the owner disputes it, the share shall become Disputed, the rider shall be notified and a ... | CS455-86, CS455-89, CS455-94, CS455-95 | #67, #69, #71 | `rides/SettlementPanel` (1), `lib/payment-settlement` (1) | Pass 2/2 |
| **FR-RD-11.4** The system shall remind the rider of a Due share 24 and 72 hours after trip completion (P-18). | CS455-76, CS455-77, CS455-89, CS455-94, CS455-95 | #55, #69, #71, #84 | `bugs/timestamp-parameters` (1) | Pass 1/1 |
| **FR-RD-11.5** The system shall block new join requests from a rider with a share Due for more than 7 days after trip completion (P-18), stating the reason. | CS455-76, CS455-77, CS455-89, CS455-94, CS455-95 | #55, #69, #71, #84 | `bugs/timestamp-parameters` (2), `lib/join-requests` (1) | Pass 3/3 |
| **FR-RD-12.1** Within 72 hours (P-16) of trip completion, the system shall allow the rider to rate the owner and each co-rider of that trip from 1 to 5. | CS455-21, CS455-22, CS455-23, CS455-39, CS455-40, +16 | #24, #36, #38, #39, #47, +5 | `rides/RatingForm` (1), `ui/ScoreField` (1), `lib/rating-rules` (1), +3 files | Pass 6/6 |

† verified by tests whose names do not cite the identifier (see `data/manual-links.json`)

| Requirement | Jira | GitHub | Tests | Result |
|---|---|---|---|---|
| **FR-RD-12.2** The system shall accept at most one rating per rater, trip and ratee, shall not allow rating oneself or anyone who was not an occupant of the same completed trip, ... | CS455-21, CS455-22, CS455-23, CS455-39, CS455-42, +8 | #24, #38, #39, #47, #71, +2 | `admin/complaints-and-enforcement` (2), `rides/RatingForm` (1), `lib/ride-ratings` (1) | Pass 4/4 |
| **FR-RD-12.3** The system shall show others only a person's average rating and number of ratings, and only once at least 3 ratings exist (P-24); individual ratings and raters ... | CS455-39, CS455-42, CS455-43, CS455-100, CS455-101, +6 | #38, #39, #75 | `rides/RatingForm` (1), `ai/chatbot-tools` (1), `lib/ride-ratings` (1) | Pass 3/3 |
| **FR-RD-12.4** When the rider gives a rating of 1 or 2, the system shall offer to open a complaint. | CS455-39, CS455-43 | #39 | `rides/RatingForm` (1) | Pass 1/1 |
| **FR-RD-13.1** Within 7 days (P-17) of a trip's completion or of a no-show, the system shall allow the rider to file a complaint against the owner or a co-rider of that trip, with ... | CS455-27, CS455-28, CS455-39, CS455-42, CS455-46, +19 | #27, #38, #42, #47, #49, +8 | `lib/complaint-rules` (3), `admin/complaints-and-enforcement` (3), `rides/[id]/complaints/route` (1), +5 files | Pass 12/12 |
| **FR-RD-13.2** When a complaint is submitted, the system shall give the rider a complaint reference and forward the complaint to complaint management. | CS455-21, CS455-22, CS455-23, CS455-39, CS455-42, +7 | #24, #38, #47, #49, #71, +2 | `rides/[id]/complaints/route` (1), `complaints/ComplaintForm` (1), `lib/complaints` (1), +1 files | Pass 4/4 |
| **FR-RD-13.3** The system shall never reveal a complainant's identity or complaint text to the person complained about or to other riders. | CS455-12, CS455-16, CS455-46, CS455-50, CS455-52, +1 | #18, #47, #49, #93 | `rides/[id]/complaints/route` (1), `complaints/ComplaintForm` (1), `lib/complaints` (1) | Pass 3/3 |
| **FR-RD-13.4** The system shall accept at most one complaint per complainant, trip and accused person, and at most 5 complaints per rider in any 24 hours (P-17). | CS455-21, CS455-22, CS455-23, CS455-39, CS455-42, +7 | #24, #38, #47, #49, #71, +2 | `rides/[id]/complaints/route` (1), `complaints/ComplaintForm` (1), `lib/complaints` (1), +1 files | Pass 4/4 |
| **FR-RD-13.5** The system shall show the rider their complaint's status --- Submitted, Under Review or Resolved --- and, once resolved, whether action was taken, without disclosing ... | CS455-46, CS455-47, CS455-50, CS455-52, CS455-53 | #42, #47, #49 | `lib/complaint-rules` (2), `(app)/complaints/pages` (1), `complaints/ComplaintForm` (1), +2 files | Pass 6/6 |
| **FR-RD-13.6** When the category Safety is selected, the system shall display the emergency contacts of FR-RD-15.3 and, if the rider's trip is in progress, offer the SOS action. | CS455-46, CS455-53 | #49 | `complaints/ComplaintForm` (1), `lib/complaint-queries` (1) | Pass 2/2 |
| **FR-RD-14.1**† The system shall notify the rider in real time of: a request accepted, rejected, expired, automatically withdrawn or needing reconfirmation; a ride cancelled; a ... | CS455-59, CS455-69, CS455-76, CS455-77 | #55, #61 | `lib/join-requests` (2), `lib/notification-realtime` (1) | Pass 3/3 |
| **FR-RD-14.2**† The system shall keep a notification centre listing notifications from the last 30 days (P-21), with an unread count and a mark-as-read action. | CS455-59, CS455-68, CS455-130 | #54, #95 | `notifications/NotificationsCentre` (1) | Pass 1/1 |
| **FR-RD-15.1** While the rider's trip is Pickup in Progress or In Transit, the system shall show an SOS action that requires a confirmation step before raising an alert. | CS455-12, CS455-16, CS455-60, CS455-61, CS455-62, +6 | #18, #56, #64, #86, #95 | `rides/[id]/sos/route` (1), `lib/sos-rules` (1), `lib/sos` (1), +1 files | Pass 4/4 |
| **FR-RD-15.2** When the rider raises an SOS, the system shall create an incident visible to administrators within 5 seconds (P-22), containing the rider, the ride and trip, all ... | CS455-12, CS455-16, CS455-21, CS455-22, CS455-23, +6 | #18, #24, #55, #64, #95, +2 | `lib/sos` (2), `trip/sos` (2), `rides/[id]/sos/route` (1), +3 files | Pass 8/8 |
| **FR-RD-15.3**† The SOS screen shall display the IIT Kanpur security emergency number and the national emergency number 112, and shall state that the platform does not dispatch ... | CS455-12, CS455-16, CS455-85 | #18, #64 | `rides/SosPanel` (1) | Pass 1/1 |
| **FR-RD-15.4** The system shall not show an SOS alert to the other members of the pool. | CS455-12, CS455-16, CS455-85, CS455-112, CS455-130 | #18, #64, #86, #95 | `lib/sos` (1), `trip/sos` (1) | Pass 2/2 |
| **FR-RD-15.5** The system shall allow the rider to mark their SOS as a false alarm; the incident shall be kept for administrator review. | CS455-60, CS455-61, CS455-62, CS455-63, CS455-64, +3 | #56, #64, #86 | `api/sos/[id]/route` (1), `rides/SosPanel` (1), `lib/sos-rules` (1), +2 files | Pass 5/5 |
| **FR-RD-16.1** The system shall show the rider their past rides with date, destination, share, settlement status and ratings given. | CS455-27, CS455-28, CS455-31 | #27, #33 | `home/RideHistoryList` (1), `lib/ride-view` (1) | Pass 2/2 |
| **FR-RD-16.2** The system shall ask for the rider's consent to use their interest tags in AI suggestions and allow the consent to be withdrawn at any time; without consent, the ... | CS455-100, CS455-101, CS455-102, CS455-103, CS455-104, +3 | #75 | `ai/chatbot-tools` (1) | Pass 1/1 |
| **FR-RD-16.3** When a rider requests account deletion, the system shall delete or anonymise their personal data within 30 days (P-25), keeping only the records needed for ... | CS455-31, CS455-86 | #33, #67 | `api/profile/delete/route` (1) | Pass 1/1 |

† verified by tests whose names do not cite the identifier (see `data/manual-links.json`)

| Requirement | Jira | GitHub | Tests | Result |
|---|---|---|---|---|
| **FR-RD-16.4**† The system shall allow the rider to download their own data in a machine-readable format. | CS455-31 | #33 | `api/profile/export/route` (1) | Pass 1/1 |
| **Rider AI agent requirements** | | | | |
| **AI-RD-01**† The rider-facing agent shall only assist with finding, comparing and understanding rides and with managing the requesting rider's own preferences. | CS455-100, CS455-101, CS455-102, CS455-103, CS455-104, +4 | #75, #95 | `ai/chatbot` (1) | Pass 1/1 |
| **AI-RD-02** The system shall allow the agent to invoke only the tools listed in §2, enforced by a server-side allowlist on every call, with the requesting rider's authorization … | CS455-100, CS455-101, CS455-102, CS455-103, CS455-104, +4 | #75, #95 | `ai/chatbot-tools` (4), `ai/chatbot` (1) | Pass 5/5 |
| **AI-RD-03** The agent may, without approval: rank rides, generate pros and cons, convert a natural-language query into search filters, summarise public profiles, and answer … | CS455-100, CS455-101, CS455-102, CS455-103, CS455-104, +3 | #75 | `ai/chatbot` (1) | Pass 1/1 |
| **AI-RD-04** A preference or tag change proposed by the agent shall take effect only after the rider confirms a displayed before/after view of the change; unconfirmed proposals … | CS455-100, CS455-101, CS455-102, CS455-103, CS455-104, +4 | #75, #95 | `api/chatbot/confirm-preference/route` (1), `ai/chatbot-tools` (1), `ai/chatbot` (1), +1 files | Pass 4/4 |
| **AI-RD-05** The agent shall never submit, withdraw or reconfirm join requests, record payments, submit ratings or complaints, post pool chat messages, raise or cancel SOS … | CS455-100, CS455-101, CS455-102, CS455-103, CS455-104, +4 | #75, #95 | `ai/chatbot-tools` (2), `ai/chatbot` (2) | Pass 4/4 |
| **AI-RD-06**† The system shall validate every agent output before display: each ride ID must refer to an open ride with an available seat, and each stated fare, time or count … | CS455-37, CS455-38 | #51 | `ai/recommendation-pipeline` (1), `ai/validate-ranking` (1) | Pass 2/2 |
| **AI-RD-07**† Agent responses shall conform to a defined output schema; a non-conforming response shall be retried once and then replaced by the fallback (AI-RD-13). | CS455-37, CS455-38, CS455-100, CS455-101, CS455-102, +5 | #51, #75 | `ai/chatbot` (1), `ai/validate-ranking` (1) | Pass 2/2 |
| **AI-RD-08**† Each agent request shall be bounded to at most 5 tool calls (P-23), 8 seconds (P-12) and 8,000 tokens (P-26), and each rider to 20 AI requests per hour (NFR-RD-11). | CS455-100, CS455-101, CS455-102, CS455-103, CS455-104, +3 | #75 | `api/chatbot/route` (1) | Pass 1/1 |
| **AI-RD-09** Rider input to the agent shall be limited to 500 characters. Rider input and every free-text field of another user's profile that reaches the agent (in particular … | CS455-100, CS455-101, CS455-102, CS455-103, CS455-104, +3 | #75 | `api/chatbot/route` (2), `ai/chatbot` (1) | Pass 3/3 |
| **AI-RD-10** The context sent to the AI service shall contain only the requesting rider's own permitted data and other users' public profile fields; tags of a rider who has not … | CS455-100, CS455-101, CS455-102, CS455-103, CS455-104, +3 | #75 | `ai/chatbot-tools` (2), `ai/chatbot` (1) | Pass 3/3 |
| **AI-RD-12** For each agent request the system shall record: request ID, pseudonymous rider ID, time, model and version, prompt template version, each tool call with arguments … | CS455-46, CS455-54, CS455-55, CS455-56, CS455-57, +9 | #50, #69, #75 | `admin/ai-decisions/page` (1), `ai/chatbot-audit` (1) | Pass 2/2 |
| **AI-RD-13**† When the AI service fails, times out, produces an invalid response, or produces no suggestion that passes validation, the system shall fall back to deterministic … | CS455-37, CS455-38, CS455-100, CS455-101, CS455-102, +5 | #51, #75 | `api/rides/recommendations/route` (1), `ai/chatbot` (1), `ai/recommendation-pipeline` (1) | Pass 3/3 |
| **AI-RD-15** The system shall let the rider report an AI suggestion or chatbot reply as wrong or inappropriate; reports shall be visible to administrators with the matching … | CS455-37, CS455-38, CS455-89 | #51, #69 | `rides/AskAiRecommendations` (2) | Pass 2/2 |
| **Rider non-functional requirements** | | | | |
| **NFR-RD-05** Seat allocation shall remain correct under concurrent access. (With $\geq$ 50 concurrent acceptance and leave operations on a ride with one available seat, repeated 100 … | CS455-27, CS455-28, CS455-113 | #27, #88, #93, #97 | `concurrency/last-seat` (1) | Pass 1/1 |
| **NFR-RD-08**† Failure of an external service shall not stop core rider functions. (With the AI service or email service unavailable: search, join, withdraw, pool chat and SOS all … | CS455-21, CS455-22, CS455-23, CS455-33, CS455-34, +8 | #24, #41, #55, #64, #85, +1 | `rides/join-requests` (2), `rides/offer-and-search` (1), `trip/sos` (1) | Pass 4/4 |
| **NFR-RD-09** A rider shall be able to read or modify only their own requests, settlements, complaints and notifications, and only the chats of pools they belong to, enforced on … | CS455-27, CS455-28, CS455-33, CS455-34, CS455-35, +3 | #27, #41, #90 | `lib/ride-view` (4), `lib/ride-status` (1) | Pass 5/5 |

† verified by tests whose names do not cite the identifier (see `data/manual-links.json`)

| Requirement | Jira | GitHub | Tests | Result |
|---|---|---|---|---|
| **NFR-RD-10** All rider input shall be validated on the server. (Every endpoint validates types, lengths and enumerations against a schema; injection test cases (SQL/NoSQL, . . . | CS455-21, CS455-22, CS455-23, CS455-46, CS455-50, +5 | #24, #47, #55, #67, #88 | `adversarial/input` (4) | Pass 4/4 |
| **NFR-RD-15** Rider screens shall work on phones and meet basic accessibility. (Usable without horizontal scrolling at 360 px width; text contrast meets WCAG 2.1 AA) | CS455-12, CS455-14, CS455-15, CS455-16, CS455-20, +3 | #16, #17, #18, #21, #23 | `contrast` (2) | Pass 2/2 |
| **NFR-RD-17** Development. Rider modules shall be developed in TypeScript and covered by automated tests. (≥ 80 % statement coverage of rider modules; continuous integration . . . | CS455-12, CS455-14, CS455-16 | #16, #18, #93 | unit coverage 87.7 % statements (gate 80 %) | Pass |
| **NFR-RD-21**† Ethical. Safety features shall not overstate what they do. (SOS text states that the platform does not dispatch emergency services (FR-RD-15.3)) | CS455-12, CS455-14, CS455-85 | #16, #64 | `app/page` (1), `rides/SosPanel` (1) | Pass 2/2 |
| **Ride Owner functional requirements** | | | | |
| **FR-RO-02.1**† The system shall allow a Ride Owner to create and update a profile containing travel preferences and optional interest tags. | CS455-31 | #33 | `api/profile/route` (1), `profile/ProfileEditor` (1) | Pass 2/2 |
| **FR-RO-03.1** The system shall allow a Ride Owner to create a ride by providing a destination, departure time window, fixed vehicle capacity, and expected total fare. | CS455-20, CS455-21, CS455-22, CS455-23, CS455-24, +7 | #21, #24, #41, #85 | `rides/offer-and-search` (1) | Pass 1/1 |
| **FR-RO-03.2** The system shall validate that the departure time is in the future, the capacity and expected total fare are valid positive values, and the Ride Owner has no . . . | CS455-20, CS455-21, CS455-22, CS455-23, CS455-24, +8 | #21, #24, #41, #67, #85 | `rides/offer-and-search` (2) | Pass 2/2 |
| **FR-RO-03.3** The system shall create a valid new ride in the Scheduled state and make it available to Riders searching for matching rides. | CS455-21, CS455-22, CS455-23, CS455-33, CS455-34, +4 | #24, #41, #85 | `rides/offer-and-search` (1) | Pass 1/1 |
| **FR-RO-04.1** The system shall allow a Ride Owner to edit the destination, departure time window, capacity, or expected total fare of an open, unlocked ride. | CS455-76, CS455-77, CS455-111 | #55, #85 | `rides/manage-ride` (1) | Pass 1/1 |
| **FR-RO-04.2** The system shall notify accepted Riders when an edit to their ride is saved. | CS455-76, CS455-77, CS455-111 | #55, #85, #93 | `rides/manage-ride` (1) | Pass 1/1 |
| **FR-RO-04.3** The system shall require the Ride Owner to confirm an edit that may affect already accepted Riders. | CS455-76, CS455-77, CS455-111 | #55, #85 | `rides/manage-ride` (1) | Pass 1/1 |
| **FR-RO-04.4** The system shall allow a Ride Owner to cancel a ride that has not started and shall notify all accepted Riders of the cancellation. | CS455-76, CS455-77, CS455-86, CS455-111 | #55, #67, #85, #93 | `rides/manage-ride` (1) | Pass 1/1 |
| **FR-RO-04.5** The system shall prevent normal edit actions after a ride is locked. A locked ride may still be cancelled before its departure window ends and before the trip . . . | CS455-33, CS455-34, CS455-35, CS455-36, CS455-39, +9 | #38, #41, #47, #55, #88, +1 | `adversarial/time-boundaries` (1), `rides/manage-ride` (1) | Pass 2/2 |
| **FR-RO-05.1** The system shall show the Ride Owner every pending request for their open ride, together with relevant Rider profile details. | CS455-12, CS455-15, CS455-33, CS455-34, CS455-35, +6 | #17, #41, #55, #85, #95 | `rides/join-requests` (1) | Pass 1/1 |
| **FR-RO-05.2** The system shall allow the Ride Owner to accept a pending request only when the ride is open and has available capacity. | CS455-33, CS455-34, CS455-35, CS455-36, CS455-45, +3 | #41, #55, #85 | `rides/join-requests` (1), `rides/seats-and-lock` (1) | Pass 2/2 |
| **FR-RO-05.3** When a request is accepted, the system shall add the Rider to the confirmed pool, update the available-seat count, and notify the Rider. | CS455-33, CS455-34, CS455-35, CS455-36, CS455-45, +3 | #41, #55, #85 | `rides/join-requests` (1) | Pass 1/1 |
| **FR-RO-05.4** The system shall allow the Ride Owner to reject a pending request without adding the Rider to the pool and shall notify the Rider of the decision. | CS455-33, CS455-34, CS455-35, CS455-36, CS455-45, +3 | #41, #55, #85 | `rides/join-requests` (1) | Pass 1/1 |
| **FR-RO-05.5** The system shall notify the Ride Owner when a new join request is submitted. | CS455-33, CS455-34, CS455-35, CS455-36, CS455-45, +3 | #41, #55, #85 | `rides/join-requests` (1) | Pass 1/1 |
| **FR-RO-06.1**† The system shall display the fixed capacity, confirmed occupants, and remaining seats for a ride. | CS455-27, CS455-28, CS455-29 | #27, #28 | `rides/RideDetails` (1), `lib/ride-view` (1) | Pass 2/2 |
| **FR-RO-06.2** The system shall allow the Ride Owner to lock a scheduled ride before P-02. | CS455-60, CS455-61, CS455-62, CS455-63, CS455-64, +4 | #55, #56, #85, #93 | `rides/seats-and-lock` (2), `lib/trip-rules` (1) | Pass 3/3 |

† verified by tests whose names do not cite the identifier (see `data/manual-links.json`)

| Requirement | Jira | GitHub | Tests | Result |
|---|---|---|---|---|
| **FR-RO-06.3** The system shall automatically lock a ride at P-02 if it has not already been locked by the Ride Owner. | CS455-76, CS455-77, CS455-111 | #55, #85 | `rides/seats-and-lock` (1) | Pass 1/1 |
| **FR-RO-06.4** The system shall prevent new join requests, request acceptances, and voluntary exits after a ride is locked. | CS455-33, CS455-34, CS455-35, CS455-36, CS455-39, +8 | #38, #41, #47, #55, #88 | `adversarial/time-boundaries` (1) | Pass 1/1 |
| **FR-RO-07.1**† The system shall display the expected total fare and the current calculated share for each confirmed occupant. | CS455-27, CS455-28, CS455-29, CS455-130 | #28, #95 | `rides/RideDetails` (2) | Pass 2/2 |
| **FR-RO-07.2** The system shall recalculate and display fare shares whenever a Rider is accepted, leaves before lock time, or is marked as a no-show. | CS455-33, CS455-34, CS455-35, CS455-36, CS455-45, +3 | #41, #55, #85 | `rides/join-requests` (1) | Pass 1/1 |
| **FR-RO-07.3**† The system shall divide the whole-ride expected total fare as evenly as possible among the owner and all active Riders; shares shall sum exactly to that fare, with … | CS455-12, CS455-16, CS455-27, CS455-28 | #18, #27 | `landing/FareCalculator` (1), `lib/ride-view` (1) | Pass 2/2 |
| **FR-RO-08.1**† The system shall provide a private text chat for the Ride Owner and accepted Riders in a confirmed pool. | CS455-33, CS455-34, CS455-35, CS455-36, CS455-45, +1 | #41, #95 | `lib/ride-chat` (2) | Pass 2/2 |
| **FR-RO-08.2**† The system shall prevent pending and rejected Riders from accessing the private ride chat. | CS455-33, CS455-34, CS455-35, CS455-36, CS455-45 | #41 | `lib/ride-chat` (1) | Pass 1/1 |
| **FR-RO-09.1** The system shall allow the Ride Owner to transition a locked ride from Scheduled to Pickup in Progress. | CS455-27, CS455-28, CS455-39, CS455-41, CS455-60, +7 | #27, #37, #56, #57, #64, +1 | `trip/trip-flow` (2), `rides/[id]/trip/route` (1), `lib/ride-view` (1), +2 files | Pass 6/6 |
| **FR-RO-09.2** The system shall notify confirmed Riders when pickup starts. | CS455-60, CS455-61, CS455-62, CS455-63, CS455-64, +3 | #57, #64, #95 | `lib/trip` (1) | Pass 1/1 |
| **FR-RO-09.3** During Pickup in Progress, the system shall display the arrival status of each confirmed Rider and allow the Ride Owner to mark a Rider as Arrived. | CS455-39, CS455-41, CS455-60, CS455-61, CS455-62, +5 | #37, #56, #57, #64, #86 | `rides/[id]/arrivals/route` (1), `lib/trip-rules` (1), `lib/trip` (1), +1 files | Pass 4/4 |
| **FR-RO-09.4** The system shall allow the Ride Owner to transition a ride from Pickup in Progress to Completed and store the final active pool and final fare shares. | CS455-27, CS455-28, CS455-29, CS455-39, CS455-41, +19 | #27, #28, #37, #56, #57, +5 | `rides/[id]/complete/route` (1), `rides/[id]/trip/route` (1), `rides/RideDetails` (1), +4 files | Pass 7/7 |
| **FR-RO-10.1** The system shall allow the Ride Owner to mark a confirmed Rider as No-Show only during Pickup in Progress and only after P-03 has elapsed. | CS455-33, CS455-34, CS455-35, CS455-36, CS455-39, +18 | #38, #41, #47, #55, #57, +5 | `lib/trip` (1), `adversarial/time-boundaries` (1), `trip/trip-flow` (1) | Pass 3/3 |
| **FR-RO-10.2** When a Rider is marked as a no-show, the system shall remove the Rider from the active pool and recalculate the fare shares for the remaining active occupants. | CS455-27, CS455-28, CS455-39, CS455-41, CS455-60, +7 | #27, #37, #56, #57, #64, +1 | `lib/trip` (3), `lib/ride-view` (1), `lib/trip-rules` (1), +1 files | Pass 6/6 |
| **FR-RO-10.3** The system shall record a no-show outcome for administrative review and the applicable rating or penalty workflow. | CS455-39, CS455-41, CS455-60, CS455-61, CS455-62, +5 | #37, #57, #64, #86 | `lib/trip` (1), `trip/trip-flow` (1) | Pass 2/2 |
| **FR-RO-11.1**† The system shall show the completed ride summary with every active Rider's final fare share and settlement status. | CS455-86, CS455-89, CS455-94, CS455-95 | #67, #69, #71 | `rides/SettlementPanel` (1) | Pass 1/1 |
| **FR-RO-11.2**† The system shall allow the Ride Owner to mark an active Rider's fare share as Settled after receiving payment outside the application. | CS455-86, CS455-89, CS455-94, CS455-95 | #67, #69, #71 | `rides/SettlementPanel` (1), `lib/payment-settlement` (1) | Pass 2/2 |
| **FR-RO-11.3**† The system shall prevent a no-show Rider from being marked as settled for the standard ride share. | CS455-86, CS455-89, CS455-94, CS455-95 | #67, #69, #71 | `rides/SettlementPanel` (1), `lib/payment-settlement` (1) | Pass 2/2 |
| **FR-RO-11.4**† The system shall send configured reminders for unsettled fare shares. | CS455-76, CS455-77, CS455-89, CS455-94, CS455-95 | #55, #69, #71, #84 | `api/cron/payment-reminders/route` (1), `bugs/timestamp-parameters` (1) | Pass 2/2 |
| **FR-RO-12.1** The system shall allow the Ride Owner to submit a rating and optional feedback for each active Rider within P-04 after completion. | CS455-21, CS455-22, CS455-23, CS455-39, CS455-42, +6 | #24, #38, #47, #71, #86 | `admin/complaints-and-enforcement` (1) | Pass 1/1 |

† verified by tests whose names do not cite the identifier (see `data/manual-links.json`)

| Requirement | Jira | GitHub | Tests | Result |
|---|---|---|---|---|
| **FR-RO-12.2** The system shall distinguish a no-show outcome from ordinary trip feedback. | CS455-21, CS455-22, CS455-23, CS455-27, CS455-28, +14 | #24, #27, #38, #47, #71, +2 | `lib/ride-view` (1), `admin/complaints-and-enforcement` (1) | Pass 2/2 |
| **FR-RO-13.2** The system shall provide an SOS action during an active ride that requests location permission and sends an urgent alert to the Operations Admin. | CS455-60, CS455-61, CS455-62, CS455-63, CS455-64, +4 | #56, #64, #86, #95 | `lib/sos-rules` (1), `lib/sos` (1), `trip/sos` (1) | Pass 3/3 |
| **FR-RO-14.1**† The system shall notify the Ride Owner of new join requests, request outcomes, saved ride changes, pool locking, and pickup start. | CS455-76, CS455-77, CS455-94, CS455-95 | #55, #71 | `lib/join-requests` (1), `lib/payment-settlement` (1) | Pass 2/2 |
| **FR-RO-14.2**† The system shall notify affected Riders of request decisions, ride changes, ride cancellation, pool locking, and pickup start. | CS455-76, CS455-77, CS455-111 | #55, #85 | `rides/manage-ride` (2), `lib/join-requests` (1) | Pass 3/3 |
| **Ride Owner non-functional requirements** | | | | |
| **NFR-RO-DEP-02** The system shall handle concurrent join requests and acceptance attempts without allowing confirmed occupants to exceed the fixed vehicle capacity. | CS455-21, CS455-22, CS455-23, CS455-76, CS455-77, +2 | #24, #55, #64, #88 | `adversarial/races` (1) | Pass 1/1 |
| **NFR-RO-DEP-04**† Once a ride is locked, the system shall reliably reject normal joins, acceptances, voluntary exits, edits, and cancellations in accordance with the configured lock … | CS455-33, CS455-34, CS455-35, CS455-36, CS455-39, +8 | #38, #41, #47, #55, #88 | `adversarial/time-boundaries` (2) | Pass 2/2 |
| **NFR-RO-SEC-01** Only the Ride Owner of a ride may accept or reject its join requests, edit or cancel it, lock it, record arrivals or no-shows, and mark settlements. | CS455-33, CS455-34, CS455-35, CS455-36, CS455-39, +12 | #37, #41, #55, #57, #85, +1 | `rides/join-requests` (1), `rides/manage-ride` (1), `rides/seats-and-lock` (1), +1 files | Pass 4/4 |
| **NFR-RO-SEC-03**† The Ride Owner's SOS location shall be transmitted only to authorized Operations Admin personnel and shall not be visible to other Riders. | CS455-85, CS455-112 | #64, #86 | `trip/sos` (3) | Pass 3/3 |
| **NFR-RO-SEC-04**† Private ride chat shall be accessible only to confirmed occupants, retained securely for 30 days after completion for audit purposes, and permanently deleted after … | CS455-33, CS455-34, CS455-35, CS455-36, CS455-45 | #41 | `lib/ride-chat` (2) | Pass 2/2 |
| **NFR-RO-SEC-06**† Complaints, no-show records, and settlement-status data shall be visible only to authorized users whose role requires access. | CS455-46, CS455-48, CS455-94, CS455-95 | #43, #71 | `rides/SettlementPanel` (1), `lib/admin-auth` (1) | Pass 2/2 |
| **NFR-RO-USE-04** Status indicators for pending, accepted, rejected, arrived, no-show, settled, and unpaid shall be visually distinct and accompanied by text labels. | CS455-27, CS455-28, CS455-29 | #28 | `rides/RideStateLabel` (1) | Pass 1/1 |
| **Operations Admin functional requirements** | | | | |
| **OA-FR-01** The system shall allow an Operations Admin to add a vehicle type to the set of vehicle types available for newly created rides. | CS455-46, CS455-47, CS455-50, CS455-51, CS455-52 | #42, #46, #47, #48 | `config/VehicleTypesEditor` (1), `lib/admin-config-rules` (1), `lib/admin-config` (1) | Pass 3/3 |
| **OA-FR-02** The system shall allow an Operations Admin to deactivate a vehicle type so that it cannot be selected for newly created rides. | CS455-46, CS455-47, CS455-50, CS455-51, CS455-52 | #42, #46, #47, #48 | `config/VehicleTypesEditor` (1), `lib/admin-config-rules` (1), `lib/admin-config` (1) | Pass 3/3 |
| **OA-FR-03** The system shall allow an Operations Admin to configure the maximum passenger capacity of each active vehicle type. | CS455-46, CS455-47, CS455-50, CS455-51, CS455-52 | #42, #46, #47, #48 | `config/VehicleTypesEditor` (1), `lib/admin-config-rules` (1), `lib/admin-config` (1) | Pass 3/3 |
| **OA-FR-05** The system shall allow an Operations Admin to add a destination hub. | CS455-46, CS455-47, CS455-50, CS455-51, CS455-52 | #42, #46, #47, #48 | `config/PlacesEditor` (1), `lib/admin-config-rules` (1), `lib/admin-config` (1) | Pass 3/3 |
| **OA-FR-06** The system shall allow an Operations Admin to modify the configuration of an active destination hub without changing the identity of the hub referenced by existing … | CS455-46, CS455-47, CS455-50, CS455-51, CS455-52 | #42, #46, #47, #48 | `lib/admin-config-rules` (2), `config/PlacesEditor` (1), `lib/admin-config` (1) | Pass 4/4 |
| **OA-FR-07** The system shall allow an Operations Admin to deactivate a destination hub so that it cannot be selected for newly created rides while preserving its use in … | CS455-46, CS455-47, CS455-50, CS455-51, CS455-52 | #42, #46, #47, #48 | `config/PlacesEditor` (1), `lib/admin-config-rules` (1), `lib/admin-config` (1) | Pass 3/3 |
| **OA-FR-11** The system shall allow an Operations Admin to configure an approximate fare range for each supported travel segment involving an external destination hub and … | CS455-46, CS455-47, CS455-50, CS455-51, CS455-52 | #42, #46, #47, #48 | `config/fares-editors` (1), `lib/admin-config-rules` (1), `lib/admin-config` (1) | Pass 3/3 |

† verified by tests whose names do not cite the identifier (see `data/manual-links.json`)

| Requirement | Jira | GitHub | Tests | Result |
|---|---|---|---|---|
| **OA-FR-16**† The system shall allow an Operations Admin to view ride information for administrative monitoring and complaint investigation. | CS455-46, CS455-53 | #49 | `admin/rides/[id]/page` (1), `lib/admin-queries` (1) | Pass 2/2 |
| **OA-FR-18** The system shall allow an authorized Operations Admin to view complaints submitted by riders against other riders. | CS455-21, CS455-22, CS455-23, CS455-39, CS455-42, +10 | #24, #38, #42, #47, #49, +3 | `admin/complaints/[id]/page` (1), `admin/complaints/[id]/route` (1), `complaints/ComplaintStatusActions` (1), +4 files | Pass 7/7 |
| **OA-FR-20** The system shall allow an authorized Operations Admin to view a reported rider's retained previous complaints. | CS455-46, CS455-53 | #49 | `admin/complaints/[id]/page` (1), `lib/admin-queries` (1) | Pass 2/2 |
| **OA-FR-21** The system shall allow an authorized Operations Admin to issue a warning to a rider and shall require a reason for the warning. | CS455-21, CS455-22, CS455-23, CS455-39, CS455-42, +10 | #24, #38, #47, #50, #71, +1 | `admin/riders/[id]/routes` (1), `riders/RiderEnforcementActions` (1), `lib/enforcement-rules` (1), +2 files | Pass 5/5 |
| **OA-FR-23** The system shall allow an authorized Operations Admin to suspend a rider account and shall require the Admin to specify a suspension duration or explicitly select … | CS455-21, CS455-22, CS455-23, CS455-39, CS455-42, +10 | #24, #38, #47, #50, #71, +1 | `admin/riders/[id]/routes` (1), `riders/RiderEnforcementActions` (1), `lib/enforcement` (1), +1 files | Pass 4/4 |
| **OA-FR-28** The system shall allow an authorized Operations Admin to manually reactivate a suspended rider. | CS455-21, CS455-22, CS455-23, CS455-39, CS455-42, +10 | #24, #38, #47, #50, #71, +1 | `admin/riders/[id]/routes` (1), `riders/RiderEnforcementActions` (1), `lib/enforcement` (1), +1 files | Pass 4/4 |
| **OA-FR-34** The system shall require explicit Operations Admin approval before executing an AI-recommended warning or suspension. | CS455-12, CS455-15, CS455-46, CS455-54, CS455-55, +2 | #17, #50, #93 | `complaints/AiComplaintReview` (1), `lib/ai-enforcement` (1) | Pass 2/2 |
| **OA-FR-36** The system shall require an Operations Admin to provide a reason when rejecting or overriding an AI enforcement recommendation. | CS455-46, CS455-54, CS455-55, CS455-56, CS455-57, +2 | #50, #71, #93 | `complaints/AiComplaintReview` (1), `lib/ai-enforcement` (1) | Pass 2/2 |
| **OA-FR-42** The system shall allow an authorized Operations Admin to view a rider's identity, account status, ride history, complaint history, warning history, and suspension … | CS455-46, CS455-53 | #49 | `admin/riders/pages` (2), `lib/admin-queries` (2) | Pass 4/4 |
| **System functional requirements (Operations Admin)** | | | | |
| **SYS-FR-04**† The system shall apply a vehicle-type capacity change only to rides created after the change and shall preserve the capacity applicable to an existing ride. | CS455-21, CS455-22, CS455-23, CS455-33, CS455-34, +4 | #24, #41, #85 | `api/rides/route` (1), `rides/offer-and-search` (1) | Pass 2/2 |
| **SYS-FR-08**† The system shall restrict rider destination selection to active configured destination hubs. | CS455-20, CS455-21, CS455-24, CS455-25, CS455-31 | #21, #33 | `api/profile/route` (1), `lib/ride-options` (1) | Pass 2/2 |
| **SYS-FR-09** The initial system configuration shall include destination hubs for the fourteen campus halls, Main Gate, the metro station, Kanpur Central railway station, Kanpur … | CS455-12, CS455-14, CS455-16, CS455-20, CS455-21, +2 | #16, #18, #21 | `app/page` (1) | Pass 1/1 |
| **SYS-FR-12** The system shall display an Operations Admin-configured external fare range as an estimate rather than as the final exact fare. | CS455-20, CS455-21, CS455-22, CS455-23, CS455-24, +6 | #21, #24, #42, #47, #48 | `config/fares-editors` (1), `rides/CreateRideForm` (1), `rides/window-and-fare` (1), +2 files | Pass 5/5 |
| **SYS-FR-14**† The system shall divide the ride's expected total fare as evenly as possible among all occupants, including the ride owner. | CS455-12, CS455-16, CS455-27, CS455-28, CS455-99, +1 | #18, #27, #74, #95 | `landing/FareCalculator` (1), `lib/ride-view` (1) | Pass 2/2 |
| **SYS-FR-15**† The system shall calculate whole-rupee fare shares that sum exactly to the ride's expected total fare, applying the deterministic rounding rule in Table T-2. | CS455-27, CS455-28, CS455-60, CS455-61, CS455-62, +4 | #27, #57, #74 | `lib/ride-view` (1), `lib/trip` (1) | Pass 2/2 |
| **SYS-FR-17** The system shall prevent an Operations Admin from modifying any ride through the Admin interface. | CS455-46, CS455-53, CS455-130 | #49, #95 | `admin/rides/[id]/page` (1), `lib/admin-queries` (1) | Pass 2/2 |
| **SYS-FR-24** The system shall prevent a suspended rider from creating or joining rides while the suspension is active. | CS455-21, CS455-22, CS455-23, CS455-39, CS455-42, +7 | #24, #38, #42, #47, #71, +1 | `lib/enforcement` (1), `lib/suspensions` (1), `admin/complaints-and-enforcement` (1) | Pass 3/3 |
| **SYS-FR-25** The system shall automatically reactivate a rider account when its temporary suspension reaches its configured expiry time. | CS455-46, CS455-47, CS455-50, CS455-52, CS455-86 | #42, #47, #67 | `lib/enforcement` (1), `lib/suspension-rule` (1), `lib/suspensions` (1) | Pass 3/3 |
| **SYS-FR-26** The system shall keep an indefinitely suspended rider suspended until an authorized Operations Admin manually reactivates the account. | CS455-46, CS455-47, CS455-50, CS455-52 | #42, #47 | `lib/enforcement` (1), `lib/suspensions` (1) | Pass 2/2 |

† verified by tests whose names do not cite the identifier (see `data/manual-links.json`)

| Requirement | Jira | GitHub | Tests | Result |
|---|---|---|---|---|
| **SYS-FR-30** The system shall display a valid AI-generated summary of a complaint to an authorized Operations Admin. | CS455-46, CS455-54, CS455-55, CS455-56, CS455-57 | #50 | `lib/complaint-ai` (1) | Pass 1/1 |
| **SYS-FR-31** The system shall preserve the original rider-submitted complaint independently of the AI-generated summary. | CS455-46, CS455-54, CS455-55, CS455-56, CS455-57 | #50 | `lib/complaint-ai` (1) | Pass 1/1 |
| **SYS-FR-32** The system shall display the AI-generated severity classification and behavioral tags for a complaint to an authorized Operations Admin when valid output is … | CS455-46, CS455-54, CS455-55, CS455-56, CS455-57 | #50 | `lib/complaint-ai` (1) | Pass 1/1 |
| **SYS-FR-33** The system shall display an AI-generated enforcement recommendation to an authorized Operations Admin when a valid recommendation is available. | CS455-46, CS455-54, CS455-55, CS455-56, CS455-57 | #50 | `lib/complaint-ai` (1) | Pass 1/1 |
| **SYS-FR-35** The system shall not execute an AI-recommended warning or suspension when an Operations Admin rejects the recommendation. | CS455-46, CS455-54, CS455-55, CS455-56, CS455-57 | #50 | `lib/ai-enforcement` (1) | Pass 1/1 |
| **SYS-FR-37** The system shall prevent the Complaint Management AI from creating, modifying, or removing a rider warning or suspension without an explicit authorized Operations … | CS455-46, CS455-54, CS455-55, CS455-56, CS455-57 | #50 | `lib/ai-enforcement` (1) | Pass 1/1 |
| **SYS-FR-38** The system shall record the complaint ID, AI recommendation, Operations Admin decision, Admin identity, timestamp, and final enforcement action for each AI … | CS455-46, CS455-53, CS455-54, CS455-55, CS455-56, +3 | #49, #50, #71 | `admin/ai-decisions/page` (1), `lib/admin-queries` (1), `lib/ai-enforcement` (1) | Pass 3/3 |
| **SYS-FR-39** The system shall authenticate a user before granting Operations Admin access. | CS455-12, CS455-15, CS455-46, CS455-48, CS455-50, +1 | #17, #43, #46 | `admin/AdminNotAllowed` (1), `lib/admin-auth` (1) | Pass 2/2 |
| **SYS-FR-43** The system shall restrict complaint, disciplinary, and AI enforcement records to authorized Operations Admins. | CS455-46, CS455-48, CS455-50, CS455-52 | #43, #46 | `lib/admin-auth` (1) | Pass 1/1 |
| **SYS-FR-45** The system shall record the authenticated Operations Admin identity, timestamp, action type, affected entity, and, when the action requires a reason, that reason … | CS455-33, CS455-34, CS455-35, CS455-36, CS455-45, +9 | #41, #43, #46, #50, #67 | `api/admin/chat-reports/route` (1), `lib/admin-actions` (1) | Pass 2/2 |
| **SYS-FR-48** The system shall report a state-changing Operations Admin action as successful only after all state changes belonging to that operation have been persisted … | CS455-46, CS455-48, CS455-50, CS455-52, CS455-54, +4 | #43, #46, #50, #67 | `lib/admin-actions` (1) | Pass 1/1 |
| **System non-functional requirements (Operations Admin)** | | | | |
| **SYS-NFR-01**† The system shall record the previous and new values for each Operations Admin configuration change. | CS455-46, CS455-50, CS455-52 | #47 | `lib/admin-api` (1), `lib/admin-config` (1) | Pass 2/2 |
| **SYS-NFR-03** The system shall enforce Operations Admin authorization on the server side for every Admin-protected operation and Admin-protected data access. | CS455-21, CS455-22, CS455-23, CS455-39, CS455-42, +8 | #24, #38, #43, #47, #71, +2 | `lib/admin-actions` (1), `admin/complaints-and-enforcement` (1) | Pass 2/2 |
| **SYS-NFR-04** The system shall reject an Operations Admin configuration update when the configuration has changed since the Admin last retrieved it. | CS455-46, CS455-47, CS455-48, CS455-50, CS455-52, +1 | #42, #43, #47, #95 | `lib/admin-actions` (1), `lib/admin-config-rules` (1), `lib/admin-config` (1) | Pass 3/3 |
| **SYS-NFR-05** The system shall clearly inform an Operations Admin when a configuration update is rejected because the configuration is stale. | CS455-46, CS455-48 | #43 | `lib/admin-actions` (1) | Pass 1/1 |
| **SYS-NFR-06** The system shall record failed Operations Admin attempts to perform suspension, reactivation, fare configuration, vehicle configuration, and destination-hub … | CS455-46, CS455-48, CS455-86 | #43, #67 | `lib/admin-actions` (1) | Pass 1/1 |
| **SYS-NFR-07** The system shall not disclose complaint, disciplinary, or AI enforcement records to a user who does not have Operations Admin authorization. | CS455-46, CS455-53, CS455-130 | #49, #95 | `admin/riders/pages` (1), `lib/admin-queries` (1) | Pass 2/2 |
| **SYS-NFR-08** The system shall associate each Operations Admin action with the authenticated Admin identity under which the action was performed. | CS455-21, CS455-22, CS455-23, CS455-39, CS455-42, +6 | #24, #38, #47, #71, #86 | `admin/complaints-and-enforcement` (1) | Pass 1/1 |
| **SYS-NFR-09** The complaint-management workflow shall remain available for manual Operations Admin review when AI processing fails, times out, produces invalid output, or … | CS455-21, CS455-22, CS455-23, CS455-39, CS455-42, +6 | #24, #38, #47, #71, #86 | `admin/complaints-and-enforcement` (1) | Pass 1/1 |

† verified by tests whose names do not cite the identifier (see `data/manual-links.json`)

| Requirement | Jira | GitHub | Tests | Result |
|---|---|---|---|---|
| **SYS-NFR-10**† AI-generated complaint summaries, severity assessments, behavioral tags, and enforcement recommendations shall not by themselves change a rider's disciplinary state. | CS455-46, CS455-54, CS455-55, CS455-56, CS455-57 | #50 | `lib/ai-enforcement` (2) | Pass 2/2 |
| **SYS-NFR-12** The system shall clearly distinguish among a successful Administrative action, a rejected action, an AI recommendation awaiting human approval, and an action ... | CS455-12, CS455-15, CS455-46, CS455-49 | #17, #44 | `home/AdminHome` (1), `lib/admin-activity` (1) | Pass 2/2 |